

The Collapse of Heterogeneity in Silicon Philosophers

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Question
Can an LLM stand in for a population of minds?
Setup

- $N = 277$ PhilPeople philosophers
- 100 PhilPapers 2020 questions
- 7 profile-conditioned LLMs
- Temperature 0; Accept $\rightarrow$ Reject coded onto $[0, 1]$
- Checked against PhilPapers 2020 ($N = 1,785$)

Silicon sampling

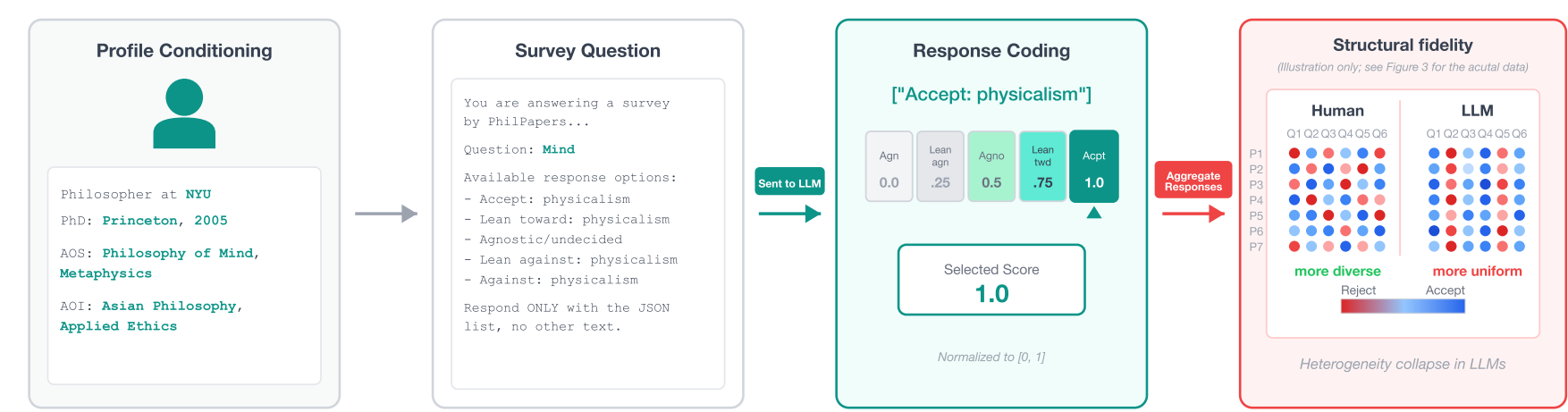


Fig. 1. Profile $\rightarrow$ question $\rightarrow$ coded stance $\rightarrow$ fidelity.

Spread collapses					
Model	N	Q	Responses	Resp%	Per-Q Var
Human	277	100	10,768	38.9%	0.053
GPT-4o	277	100	16,670	60.2%	0.014
GPT-5.1	276 [†]	100	16,526	59.9%	0.014
Claude-Sonnet-4.5	277	100	16,612	60.0%	0.026
Llama-3.1-8B	277	100	19,144	69.1%	0.026
Llama-3.1-8B (FT)	277	100	18,413	66.5%	0.019
Mistral-7B	277	100	18,764	67.7%	0.028
Qwen-3-4B	277	100	23,077	83.3%	0.016

Paper Table 1. Humans are 1.9–3.9 $\times$ more variable. [†]One silicon philosopher returned empty results.

Closest still not close			
Model	KL Div. ($\downarrow$)	JS Div. ($\downarrow$)	Pearson r ($\uparrow$)
Claude-Sonnet-4.5	0.052	0.010	0.495
Llama-3.1-8B	0.055	0.015	0.268
Llama-3.1-8B (FT)	0.291	0.087	0.272
Mistral-7B	0.653	0.178	0.264
Qwen-3-4B	0.907	0.160	0.291
GPT-5.1	0.974	0.142	0.422
GPT-4o	1.012	0.145	0.416

Paper Table 2. Flattened 277 $\times$ 100 matrices vs. human.

Normative domains fail most				
Rank	Domain	RMSE	H.Var	N
1	Aesthetics	0.216	0.014	1
2	Philosophy of Language	0.257	0.036	7
3	History of Philosophy	0.271	0.031	4
4	Political & Social Phil.	0.278	0.044	7
5	Phil. Methodology	0.279	0.064	6
6	Philosophy of Mind	0.284	0.044	14
7	Metaphysics	0.295	0.040	17
8	Decision Theory	0.297	0.032	3
9	Epistemology	0.310	0.053	10
10	Philosophy of Science	0.323	0.032	4
11	Logic & Formal Phil.	0.349	0.036	5
12	Ethics & Moral Phil.	0.362	0.082	10
13	Applied Ethics	0.406	0.100	8
14	Philosophy of Religion	0.412	0.097	4

Paper Table 5. RMSE tracks human variance ($r = 0.75$).



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Take a picture for the paper and data explorer.

A social world model is adequate only if it matches the distribution of human minds.

Silicon philosophers fail this test: they are 1.9–3.9 $\times$ too uniform, and they disagree along the wrong axes.

Human vs. Claude

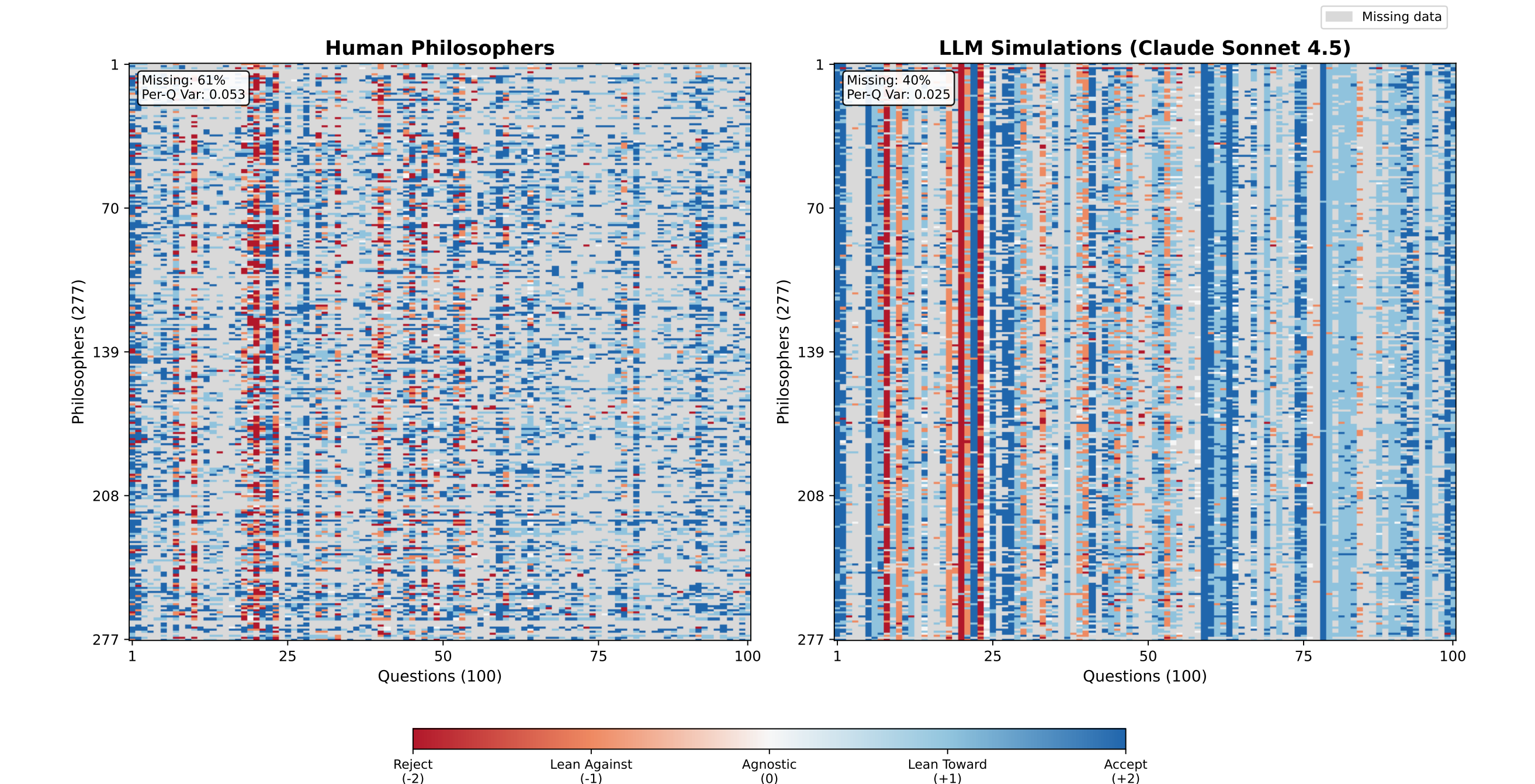


Fig. 3a. Each cell is one philosopher on one question. Humans speckled; Claude forms vertical bands.

Wrong axes				
Source	Var(6)	Elem r	Load r	Q. Overlap
Human ($N = 277$)	71.1%	—	—	—
Claude-Sonnet-4.5	72.3%	0.245**	0.023	0.8/5
GPT-4o	83.0%	0.103**	0.110	0.8/5
GPT-5.1	81.6%	0.099**	−0.137	0.7/5
Llama-3.1-8B (FT)	61.2%	0.065*	−0.063	0.7/5
Mistral-7B	73.8%	0.027	−0.041	0.7/5
Qwen-3-4B	67.8%	0.027	0.045	0.8/5
Llama-3.1-8B	62.4%	0.016	0.026	0.7/5

Paper Table 6. GPT models dump more variance into 6 axes. Best loading alignment is still weak.

Specialist Effect	Ground Truth ($N = 277$)		LLM Predictions	
	Diff	Sig	Avg Diff	Sig Models
Phil. Biology $\rightarrow$ Personal identity: biological	+11.4 pp	n.s.	+43 pp	4/7***
Phil. Biology $\rightarrow$ Personal identity: psychological	+4.1 pp	n.s.	−65.7 pp	3/7***
Ancient Phil. $\rightarrow$ Practical reason: Aristotelian	+1.9 pp	n.s.	+68.9 pp	7/7***

Paper Table 3. Ground truth n.s.; LLMs invent the link.

Question structure				
Model	Elem r	RV	KL	JS
Sonnet-4.5	0.182**	0.318	0.188	0.048
GPT-5.1	0.089**	0.307	0.468	0.110
GPT-4o	0.088**	0.318	0.485	0.114
Llama-3.1-8B (FT)	0.044*	0.224	0.176	0.046
Mistral-7B	0.043*	0.267	0.387	0.088
Qwen-3-4B	0.024	0.321	0.670	0.138
Llama-3.1-8B	0.020	0.234	0.195	0.046

Paper Table 7. 100 $\times$ 100 question-correlation matrices. ** $p < 0.01$, * $p < 0.05$.

Model	DPO does not restore it				
	Entr.	KL	Elem r	C.KL	C.JS
Llama Base	0.737	0.07	0.020	0.195	0.046
Llama DPO	0.794	0.40	0.044*	0.176	0.046